

RAMDANI KHALED

AI Systems Builder | Analytics Engineer

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PROFESSIONAL SUMMARY

Final-year Computer Science student at Université Badji Mokhtar – Annaba, specializing in Analytics Engineering and AI. Currently building AI-powered voice assistants and multi-agent ML systems. Machine learning builder driven by one goal: build AI systems with real-world impact.

EXPERIENCE

Founding AI Engineer

Jan 2026 – Present

Avanova – AI-Powered Business Assistants | Annaba, Algeria

- Architected and built the full AI pipeline for the Avanova Smart Avatar Assistant: STT (Whisper) → RAG (ChromaDB) → LLM (qwen3) → TTS (Supertonic) with real-time WebSocket streaming
- Built a 3D avatar system using Three.js + GLB models with real-time lip sync driven by audio amplitude and frequency analysis across custom morph targets
- Integrated GPU acceleration (CUDA, onnxruntime-gpu) achieving sub-1-second end-to-end latency on RTX 3060 — down from 10+ seconds on CPU
- Implemented multilingual support (English, French, Arabic) with automatic language detection and per-language TTS voice switching
- Designed the full cloud architecture: FastAPI backend, Chroma vector DB with per-client namespace isolation, auto VAD, mic pause/resume, error handling, and auto-reconnect
- Delivered a production-ready B2B MVP — a voice avatar assistant for hotel reception desks, retail entrances, and office environments — in under 2 months, solo

AI Systems Intern

Jan 2026 – Present

ANEM (Agence Nationale de l'Emploi) – Algerian National Employment Agency

- Architecting an employment intelligence platform using a multi-agent ML architecture on large-scale NoSQL Big Data
- Designed a four-stream NoSQL pipeline spanning Profiles, Offers, Placements, and References using MongoDB and Elasticsearch
- Built a KNN-based recommendation engine for skill gap analysis and job-candidate matching
- Implemented dynamic weight optimization using supervised learning trained on historical placement data

PROJECTS

Avanova Smart Avatar Assistant

2026

Python · FastAPI · Three.js · ChromaDB · Whisper · Supertonic · qwen3 · CUDA · React

- Full-stack AI voice assistant with 3D avatar: speaks, listens, and responds in real time using RAG-grounded LLM responses
- Real-time lip sync via audio frequency analysis driving GLB morph targets (Jaw_Open, mounth_open, Eye_Blink) at 30 FPS
- Sub-1-second TTS on GPU; fully offline after first model download; trilingual (EN/FR/AR)
- Deployed as B2B SaaS MVP for hotel lobbies, retail entrances, and office receptions

ARIA – Adaptive Reasoning & Intelligence Architecture

In Development

Python · LangGraph · Ollama · ChromaDB · Streamlit

- Multi-brain AI agent (Planner, Advisor, Meta-Evaluator) with human approval checkpoint before execution
- Three-layer memory: Working Memory, Episodic Memory (ChromaDB), Strategic Memory (NetworkX knowledge graph)
- Privacy-first local inference via Ollama with optional cloud fallback; SHA-256 cryptographic audit trails per step

AI Engine – Employability Scoring System

Active · Internship

MongoDB · Elasticsearch · scikit-learn · Python

- Multi-agent ML system for employment analytics processing large-scale workforce records
- Per-strategy Logistic Regression models with feature importances fed back as dynamic weight overrides

Wyag Lite – Git Reimplementation in Python

Complete

Python · Systems Programming · Git Internals

- Rebuilt Git internals from scratch in pure Python with zero external dependencies; implements commits, trees, refs, tags, and log

TECHNICAL SKILLS

AI & Agents: Multi-agent architecture, LLM orchestration, LangGraph, RAG, vector stores, local inference (Ollama), agentic pipelines, real-time voice AI

Voice AI: Whisper STT, Supertonic TTS, VAD, WebSocket streaming, lip sync, avatar animation (Three.js, GLB morph targets)

Machine Learning: scikit-learn, NumPy, Pandas, OpenCV, MediaPipe, supervised learning, KNN

Data Engineering: MongoDB, Elasticsearch, ChromaDB, NoSQL design, SQL

GPU & Inference: CUDA, onnxruntime-gpu, ctranslate2, NVIDIA RTX optimization

Development: Python (primary), JavaScript, React, FastAPI, REST APIs, Git, C, Java (basic)

EDUCATION

Bachelor of Science in Computer Science

Expected: 2026

Université Badji Mokhtar – Annaba, Algeria | Member, AI & Robotics Association of Annaba Wilaya

LANGUAGES & AFFILIATIONS

Arabic (Native) | English B1-B2 (3 years university instruction) | Member, AI & Robotics Association of Annaba Wilaya (government-recognized)